Space restriction to the OBS virtual space

generated by `space_restriction.py`

In the CF-F12 construction of Yanai the virtual space splits into the OBS virtual space and the CABS+∞ remainder. We restrict the virtual space to OBS-vir: every virtual index is an OBS-vir index, and the amplitude t_{ij}^{ab} vanishes when both of its indices are OBS-vir. In the many-body expansions of `many_body_equations.tex` this means:

- a virtual index contracted with another tensor is OK;
- a term is zero iff both virtual indices of *one and the same* t_{ij}^{ab} amplitude (or t^*) sit on the operator $\hat{E}$ – the term then carries the vanishing amplitude. Two free virtual indices stemming from *different* amplitudes do not kill the term.

Each block is shown first as the original expansion with the vanishing terms underlined, then as the restricted expansion gathering the nonzero terms. Blocks still containing operators of rank higher than two are marked; their cumulant expansion is done later.

1 First-order commutators

$\left[\hat{h},\, \hat{T}_1\right]$ **(2 terms, 0 vanish)**

$$\left[\hat{h},\, \hat{T}_1\right] = -h_p^i t_i^a \hat{E}_p^a + h_p^a t_i^a \hat{E}_i^p \tag{1}$$

restricted:

$$\left[\hat{h},\, \hat{T}_1\right] = -h_p^i t_i^a \hat{E}_p^a + h_p^a t_i^a \hat{E}_i^p \tag{2}$$

$\left[\hat{h},\, \hat{T}_2\right]$ **(2 terms, 1 vanish)**

$$\left[\hat{h},\, \hat{T}_2\right] = \underline{-h_p^j t_{ij}^{ab} \hat{E}_{ip}^{ab}} + h_p^b t_{ij}^{ab} \hat{E}_{ij}^{ap} \tag{3}$$

restricted:

$$\left[\hat{h},\, \hat{T}_2\right] = h_p^b t_{ij}^{ab} \hat{E}_{ij}^{ap} \tag{4}$$

$\left[\hat{V}, \hat{T}_1\right]$ **(2 terms, 0 vanish)**

$$\left[\hat{V}, \hat{T}_1\right] = -g^{ip}_{qr} t^a_i \hat{E}^{ap}_{qr} + g^{ar}_{pq} t^a_i \hat{E}^{pq}_{ir} \tag{5}$$

restricted:

$$\left[\hat{V}, \hat{T}_1\right] = -g^{ip}_{qr} t^a_i \hat{E}^{ap}_{qr} + g^{ar}_{pq} t^a_i \hat{E}^{pq}_{ir} \tag{6}$$

$\left[\hat{V}, \hat{T}_2\right]$ **(4 terms, 2 vanish)**

$$\left[\hat{V}, \hat{T}_2\right] = \underline{-\tfrac{1}{2}\, g^{ij}_{pq} t^{ab}_{ij} \hat{E}^{ab}_{pq}} + \tfrac{1}{2}\, g^{ab}_{pq} t^{ab}_{ij} \hat{E}^{pq}_{ij} \tag{7}$$

$$\underline{-g^{jp}_{qr} t^{ab}_{ij} \hat{E}^{abp}_{iqr}} + g^{br}_{pq} t^{ab}_{ij} \hat{E}^{apq}_{ijr} \tag{8}$$

restricted:

$$\left[\hat{V}, \hat{T}_2\right] = \tfrac{1}{2}\, g^{ab}_{pq} t^{ab}_{ij} \hat{E}^{pq}_{ij} + g^{br}_{pq} t^{ab}_{ij} \hat{E}^{apq}_{ijr} \quad \text{(three-body terms)} \tag{9}$$

2 Second order: Fock blocks

$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_1\right]$ **(1 terms, 0 vanish)**

$$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_1\right] = -2\, f^b_j t^b_i t^a_j \hat{E}^a_i \tag{10}$$

restricted:

$$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_1\right] = -2\, f^b_j t^b_i t^a_j \hat{E}^a_i \tag{11}$$

$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_2\right]$ **(2 terms, 1 vanish)**

$$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_2\right] = -f^c_k t^a_k t^{bc}_{ji} \hat{E}^{ab}_{ij} \tag{12}$$

$$\underline{-f^c_k t^c_j t^{ab}_{ik} \hat{E}^{ab}_{ij}} \tag{13}$$

restricted:

$$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_2\right] = -f^c_k t^a_k t^{bc}_{ji} \hat{E}^{ab}_{ij} \tag{14}$$

$\left[\left[\hat{F}, \hat{T}_2\right], \hat{T}_1\right]$ **(2 terms, 1 vanish)**

$$\left[\left[\hat{F}, \hat{T}_2\right], \hat{T}_1\right] = \underline{-f^c_k t^{ab}_{ik} t^c_j \hat{E}^{ab}_{ij}} \tag{15}$$

$$-\, f^c_k t^{bc}_{ji} t^a_k \hat{E}^{ab}_{ij} \tag{16}$$

restricted:

$$\left[\left[\hat{F}, \hat{T}_2\right], \hat{T}_1\right] = -f^c_k t^{bc}_{ji} t^a_k \hat{E}^{ab}_{ij} \tag{17}$$

$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right]$ **(1 terms, 1 vanish)**

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right] = \underline{-2\,f^d_l t^{cd}_{kj} t^{ab}_{il}\,\hat{E}^{abc}_{ijk}} \tag{18}$$

restricted:

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right] = 0 \tag{19}$$

$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_1\right]$ **(4 terms, 0 vanish)**

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_1\right] = f^i_j t^b_j t^a_i \hat{E}^a_b - f^b_p t^b_i t^a_i \hat{E}^a_p \tag{20}$$

$$+\, f^a_b t^a_i t^b_j \hat{E}^i_j - f^j_p t^a_j t^a_i \hat{E}^p_i \tag{21}$$

restricted:

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_1\right] = f^i_j t^b_j t^a_i \hat{E}^a_b - f^b_p t^b_i t^a_i \hat{E}^a_p \tag{22}$$

$$+\, f^a_b t^a_i t^b_j \hat{E}^i_j - f^j_p t^a_j t^a_i \hat{E}^p_i \tag{23}$$

$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right]$ **(4 terms, 2 vanish)**

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right] = \underline{f^j_k t^c_k t^{ab}_{ji} \hat{E}^{ab}_{ci}} \tag{24}$$

$$\underline{-f^c_p t^c_j t^{ab}_{ij} \hat{E}^{ab}_{ip}} + f^b_c t^c_i t^{ab}_{jk} \hat{E}^{ai}_{jk} \tag{25}$$

$$-\, f^k_p t^b_k t^{ab}_{ij} \hat{E}^{ap}_{ij} \tag{26}$$

restricted:

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right] = f^b_c t^c_i t^{ab}_{jk} \hat{E}^{ai}_{jk} \tag{27}$$

$$-\, f^k_p t^b_k t^{ab}_{ij} \hat{E}^{ap}_{ij} \tag{28}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right]$ **(8 terms, 2 vanish)**

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right] = \underline{f^j_k t^{bc}_{ki} t^a_j \hat{E}^{ai}_{bc}} \tag{29}$$

$$-\, f^c_p t^{bc}_{ji} t^a_j \hat{E}^{ai}_{bp} - f^c_p t^{bc}_{ij} t^a_j \hat{E}^{ai}_{pb} \tag{30}$$

$$\underline{+f^i_p t^{bc}_{ji} t^a_j \hat{E}^{ap}_{bc}} + f^b_c t^{ab}_{ij} t^c_k \hat{E}^{ij}_{ak} \tag{31}$$

$$+\, f^a_p t^{ab}_{ji} t^b_k \hat{E}^{ij}_{kp} - f^k_p t^{ab}_{ik} t^b_j \hat{E}^{ip}_{aj} - f^k_p t^{ab}_{ki} t^b_j \hat{E}^{ip}_{ja} \tag{32}$$

restricted:

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right] = -f^c_p t^{bc}_{ji} t^a_j \hat{E}^{ai}_{bp} \tag{33}$$

$$- f^c_p t^{bc}_{ij} t^a_j \hat{E}^{ai}_{pb} + f^b_c t^{ab}_{ij} t^c_k \hat{E}^{ij}_{ak} + f^a_p t^{ab}_{ji} t^b_k \hat{E}^{ij}_{kp} \tag{34}$$

$$- f^k_p t^{ab}_{ik} t^b_j \hat{E}^{ip}_{aj} - f^k_p t^{ab}_{ki} t^b_j \hat{E}^{ip}_{ja} \tag{35}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right]$ **(12 terms, 6 vanish)**

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right] = \underline{f^i_j t^{cd}_{jk} t^{ab}_{ik} \hat{E}^{ab}_{cd}} \tag{36}$$

$$\underline{-f^d_p t^{cd}_{ij} t^{ab}_{ij} \hat{E}^{ab}_{cp}} + f^a_b t^{ac}_{ij} t^{bc}_{kl} \hat{E}^{ij}_{kl} \tag{37}$$

$$- f^l_p t^{ab}_{il} t^{ab}_{jk} \hat{E}^{ip}_{jk} \underline{+ f^k_l t^{cd}_{li} t^{ab}_{kj} \hat{E}^{abi}_{cjd}} \tag{38}$$

$$\underline{-f^d_p t^{cd}_{ki} t^{ab}_{kj} \hat{E}^{abi}_{cjp} - f^d_p t^{cd}_{ik} t^{ab}_{jk} \hat{E}^{abi}_{jpc}} \tag{39}$$

$$\underline{+f^j_p t^{cd}_{kj} t^{ab}_{ki} \hat{E}^{abp}_{cid}} + f^c_d t^{bd}_{ij} t^{ac}_{kl} \hat{E}^{aij}_{kbl} \tag{40}$$

$$+ f^b_p t^{bc}_{ji} t^{ac}_{kl} \hat{E}^{aij}_{klp} - f^l_p t^{bc}_{il} t^{ac}_{jk} \hat{E}^{aip}_{jbk} \tag{41}$$

$$- f^l_p t^{bc}_{li} t^{ac}_{jk} \hat{E}^{aip}_{jkb} \tag{42}$$

restricted:

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right] = f^a_b t^{ac}_{ij} t^{bc}_{kl} \hat{E}^{ij}_{kl} \tag{43}$$

$$- f^l_p t^{ab}_{il} t^{ab}_{jk} \hat{E}^{ip}_{jk} + f^c_d t^{bd}_{ij} t^{ac}_{kl} \hat{E}^{aij}_{kbl} \tag{44}$$

$$+ f^b_p t^{bc}_{ji} t^{ac}_{kl} \hat{E}^{aij}_{klp} - f^l_p t^{bc}_{il} t^{ac}_{jk} \hat{E}^{aip}_{jbk} \tag{45}$$

$$- f^l_p t^{bc}_{li} t^{ac}_{jk} \hat{E}^{aip}_{jkb} \quad \text{(three-body terms)} \tag{46}$$